Challenge case studies

CPPUG Power Platform Hackathon | Saturday, October 10, 2026

Pick one of these prepared challenges, or choose Open Build and bring your own idea. Each one fits a four-hour build. Build the starter version first. Add stretch ideas only when the starter works. Judges score the core features you list on your team summary card, so pick 3 you can finish.

1. Inventory Management

Scenario	A small nonprofit lends out event gear (tables, tents, speakers). Nobody knows what is on the shelf or who has it.
Users	Volunteers who check gear out, and the coordinator who reorders it.
Starter build (MVP)	Add, edit, and view items with quantity and location. Check an item out and back in.
Stretch ideas	Barcode scan, a low-stock email alert, a reorder approval.
Suggested data	Items (name, category, quantity, location, minimum stock). Checkouts (item, person, date out, date back).
Likely tools	Power Apps, SharePoint list or Dataverse, Power Automate

2. Time Off Management

Scenario	A team of 15 asks for vacation by email. Requests get lost and managers double-book coverage.
Users	Employees who ask for time off, and the manager who approves it.
Starter build (MVP)	Submit a request (who, dates, reason). See my requests and their status.
Stretch ideas	One-step approval with Outlook or Teams, a team calendar view, remaining-days balance.
Suggested data	Requests (employee, start, end, type, reason, status, approver).
Likely tools	Power Apps, Power Automate (approvals), Outlook, SharePoint list or Dataverse

3. Inspection System

Scenario	Site supervisors fill in paper safety checklists. The office types them up days later, if at all.
Users	Field inspectors on a phone, and the safety lead who reviews results.
Starter build (MVP)	A phone-friendly checklist with pass or fail per item, notes, and an overall status.
Stretch ideas	Photo capture, a generated PDF report, an emailed summary, follow-up tasks for failed items.
Suggested data	Inspections (site, inspector, date, status). Inspection items (question, result, note, photo).
Likely tools	Power Apps (phone layout), Power Automate, SharePoint or Dataverse

4. Expense Tracking

Scenario	Staff send receipt photos by text message. Finance rebuilds the expense claims in a spreadsheet every month.
Users	Staff who spend money, the approver, and the finance person.
Starter build (MVP)	Submit an expense with date, category, amount, and a receipt. List my expenses.
Stretch ideas	Approval flow, AI Builder receipt reading, a Power BI summary by category and month.
Suggested data	Expenses (person, date, category, amount, receipt, status).
Likely tools	Power Apps, Power Automate, AI Builder, Power BI

5. Room or Equipment Booking

Scenario	A shared office has 3 meeting rooms and 2 projectors. People book them on a whiteboard that gets erased.
Users	Anyone who needs a room or device, and the office admin.
Starter build (MVP)	See what is free and request a time slot. No double bookings.
Stretch ideas	Confirmation email, a sign-out log for equipment, a Teams tab.
Suggested data	Resources (name, type, capacity). Bookings (resource, person, start, end).
Likely tools	Power Apps, SharePoint list or Dataverse, Outlook

6. Employee Kudos

Scenario	Managers want people to thank each other more, but thanks get lost in chat threads.
Users	Every employee, and the people team who runs recognition.
Starter build (MVP)	Send kudos to a colleague with a short message and a value or badge. See recent kudos.
Stretch ideas	Post each kudos to a Teams channel, a badge gallery, a monthly leaderboard in Power BI.
Suggested data	Kudos (from, to, message, badge, date).
Likely tools	Power Apps, Power Automate, Teams, Power BI

7. Open Build

Scenario	Bring your own problem. Something from work, your community, or your life.
Users	You decide. Name a real user.
Starter build (MVP)	One main task a new user can finish without help.
Stretch ideas	Anything useful, clever, or delightfully strange.
Suggested data	You decide. Sketch it in the first 30 minutes.
Likely tools	Any Power Platform parts, as long as Power Platform is central

Tip. Start with sample data. Ten realistic rows make a better demo than an empty app. AI tools, Microsoft Learn, and search are all allowed. Tell the judges what you used.